

Landmark Symmetry and Cognitive Map Formation in Predictive Recurrent Neural Networks

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Project Report: CS ISM

Abstract

Predictive learning has been proposed as a general mechanism for forming cognitive maps in the hippocampus: a recurrent neural network trained to predict future observations from the statistics of an animal's movement will develop place-cell-like spatial representations [3]. A natural question is how this framework behaves when the environment provides ambiguous sensory information. We extend the Levenstein et al. predictive RNN (pRNN) framework to ask: does rotational landmark symmetry cause map formation to fail?

We train predictive RNNs in arenas with C1, C2, and C4 rotational landmark symmetry and evaluate the resulting representations using four metrics designed to detect the forms of failure that standard spatial RSA cannot solve: Permutation-Aware Alignment (PAA), Rotational Autocorrelation (RA), C2 Contrast, and a Symmetry Collapse Index (SCI). We find that map formation does not fail under C4 symmetry. Instead, a stable spatial map forms, but is geometrically distorted: units develop rate maps that are similar at rotationally equivalent positions, and decoding precision degrades monotonically with symmetry order (RA: $S1 = 0.097$, $S2 = 0.114$, $S4 = 0.217$). Global orientational degeneracy is prevented by the head direction (HD) input: PAA gain remains near zero for all conditions ($S4 < 0.05$). Removing HD input causes map formation to collapse (sRSA: $0.615 \rightarrow 0.260$), confirming that HD is necessary for global stability. Through these studies we can say that symmetry does not cause the cognitive map formation to fail but instead deforms the geometry of the formed representation.

Introduction

Levenstein et al. (2024) showed that a recurrent network trained on sequential predictive loss spontaneously develops spatially structured representations, but validated this exclusively in an L-shaped arena chosen to ensure every position has a unique visual fingerprint. However, Most rodent electrophysiology experiments use square arenas. Square arenas are rotationally symmetric. The condition Levenstein's validation depends on (global observational discriminability) does not hold there. We ask what the Levenstein framework actually does under these conditions. Not what the hippocampus does. What the model does. The answer has direct consequences for how the framework should be used, what its representations are grounded in, and what any extension to more naturalistic environments should expect to find. (The hippocampus is a different question, and we can only speculate about that based on the model's behavior.)

The head-direction (HD) system tracks global orientation through path integration while landmark anchoring corrects drift via visual cues [7, 1], but symmetric landmarks force these two mechanisms into direct conflict: the same landmark view appears at multiple rotations, so the HD state and the landmark signal become discordant. The double-rotation literature documents that place cells increase their number of firing fields in symmetric arenas [5, 1], with multi-field fraction scaling with symmetry order, and these multi-field patterns emerge spontaneously in the hippocampus without explicit instruction [2].

This project asks whether landmark symmetry causes global map degeneracy (where different seeds learn incompatible global orientations), local representational folding (where symmetric positions collapse within a single run), or both. A standard metric (seed-to-seed

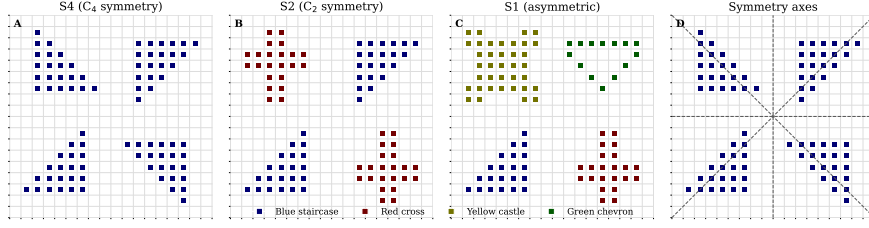


Figure 1: Arena layouts by symmetry. S4 has C_4 symmetry (four-fold rotational equiv.); S2 has C_2 symmetry (two-fold); S1 is asymmetric. Landmark shapes and positions are matched for observational discriminability across conditions.

RSA) cannot answer this question because row permutations preserve pairwise distances, making it blind to both global rotations and local folding. We introduce three metrics that separate identifiability from entanglement: PAA and RA for global vs local, C2 Contrast for group structure, and use them to test predictive networks trained on arenas with graded rotational symmetry. We show that head-direction input prevents global degeneracy while local precision degrades monotonically with symmetry order, and that this partial-folding pattern explains multi-field place cell emergence, predicts specific lesion deficits, and uncovers why sRSA fails under observation aliasing.

The theoretical link between predictive learning and spatial representation comes from the successor representation (SR) [6]. The SR encodes each state as the discounted expected future occupancy of all other states: $M(s, s') = \mathbb{E} [\sum_{t=0}^{\infty} \gamma^t \mathbf{1}(s_t = s') \mid s_0 = s]$. Under a random policy in an open environment, the SR matrix has a structure similar to a Gaussian centered at each state, producing place-cell-like activity profiles [6]. Critically, the SR geometry is shaped by the transition structure of the environment, not just its physical geometry. This means that environments with rotational symmetry will have SR matrices in which multiple distinct spatial positions are predicted to share the same future state distribution: the SR entries for these positions will be identical or nearly so. Whether a predictive RNN can resolve this ambiguity, and if not, in what way its representation fails, is one of the central questions I want to address.

Methods

The arena is an 18×18 MiniGrid (a grid-world environment where agents move along discrete rows and columns) with 324 positions. All conditions are matched on observational discriminability:

$$\text{ODI} = 1 - \frac{H(X \mid O)}{H(X)} = 0.336.$$

S4 has C_4 -equivalent observations under 90° rotation (equivalence class size $H_2 = 4.0$), S2 has C_2 -equivalent observations under 180° rotation ($H_2 = 2.0$), and S1 has no designed equivalence ($H_2 = 1.0$). The number of seeds is S4 $n = 7$, S2 $n = 5$, S1 $n = 5$.

The model is a NormReLU pRNN (layer-normalized ReLU nonlinearity: $\text{NormReLU}(\mathbf{x}) = \text{ReLU}(\mathbf{x}) / \|\text{ReLU}(\mathbf{x})\|_2$) with $N = 500$ units. At time t , the hidden state updates as:

$$h_t = \text{NormReLU}(W_h h_{t-1} + W_o o_t + W_a a_t + W_d d_t),$$

where o_t is the egocentric observation, a_t is the action, and d_t is a 5-dimensional SpeedHD vector (speed + 4-way one-hot head direction). Training minimises multi-step prediction loss for $k = 5$ steps:

$$\mathcal{L} = \sum_{j=1}^k \|\hat{o}_{t+j} - o_{t+j}\|_2^2,$$

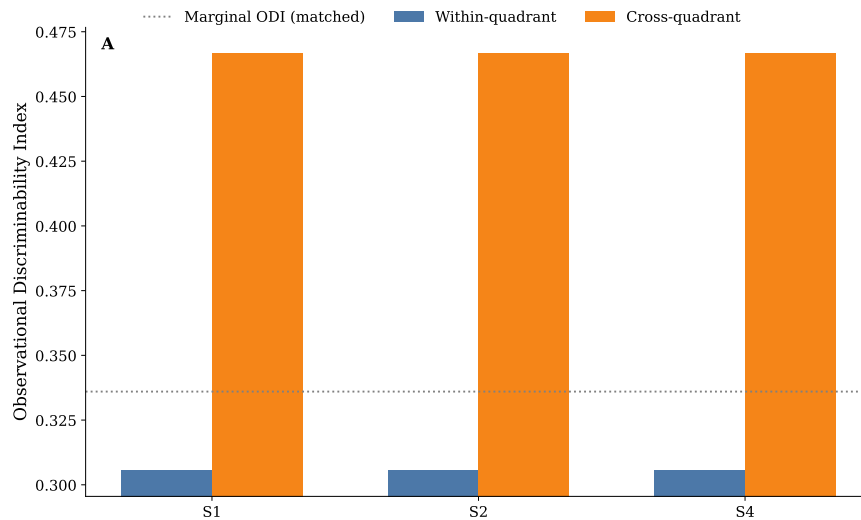


Figure 2: Observational discriminability (ODI) decomposition by condition. Within-condition and cross-condition pairwise distances confirm that all three conditions are matched on $ODI = 0.336$ despite different symmetry orders.



Figure 3: Why sRSA is invariant to permutations. Row permutations of the neural population matrix leave all pairwise distances unchanged, so rotated maps yield identical sRSA values even when they represent opposite global orientations.

over 80,000 training steps. A total of 17 seeds were run across all conditions (S4 $n = 7$, S2 $n = 5$, S1 $n = 5$).

Three metrics address this blindness. PAA detects when rotated alignment between seeds beats identity alignment, revealing orientational divergence. RA measures whether units respond identically at rotationally equivalent positions, capturing unit-level aliasing. C2 Contrast tests whether the population code encodes group structure. PAA, RA, and C2 Contrast separate identifiability, entanglement, and population structure.

Why sRSA cannot detect the failure modes this project measures

Seed-to-seed RSA correlates pairwise neural distances with pairwise spatial distances. However, this metric has a critical vulnerability: row permutations of the neural population vector preserve all pairwise distances, so rotated maps yield identical sRSA values even though they represent opposite global orientations.

Figure 3 Here we can see this more clearly: two fundamentally different failure modes (global rotation versus local folding) yield identical sRSA despite producing qualitatively different neural codes. This blindness matters empirically: sRSA is near-equal for S2 and S4 (0.72 vs 0.67), yet RA differs by nearly 2times (0.114 vs 0.217). A researcher relying on sRSA alone would conclude that C2 and C4 symmetry impose equivalent representational costs. They do not.

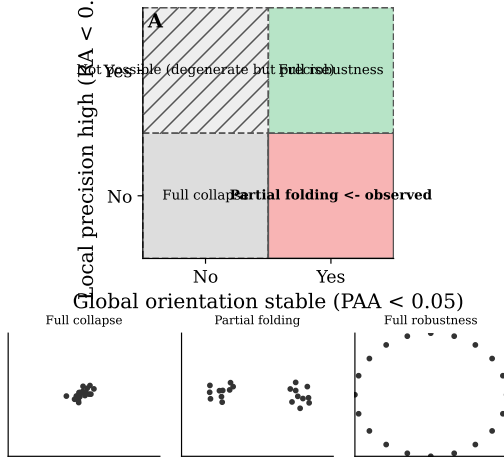


Figure 4: Mechanistically distinct outcomes. Left: collapse. Center: partial folding (observed). Right: null case.

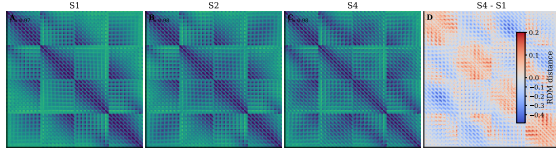


Figure 6: Cross-seed RDM (Representational Dissimilarity Matrix) alignment. S4, S2, and S1 all show strong RSA alignment across seeds.

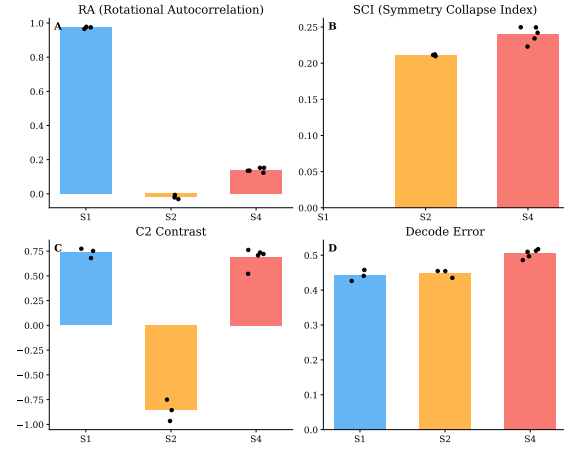


Figure 5: Final metric comparison. PAA (global), RA (aliasing), SCI (compression), and C2 Contrast (grouping).

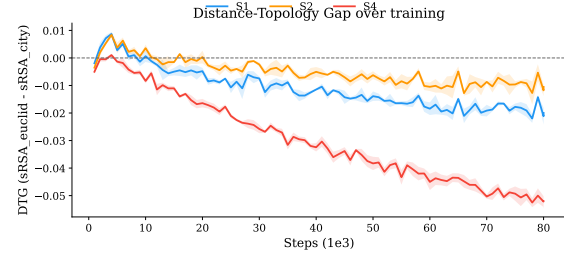


Figure 7: DTG (Distance-Topology Gap, Δ_{TG}). Gap widens in S4, indicating path-dependence.

What I expect to find

Does global orientation remain stable across seeds? Does local precision stay high? This produces three mechanistically distinct failure modes: (1) global degeneracy and local collapse, (2) global stability but local folding (the case we observe), or (3) full robustness.

These four metrics showcase the following: global structure is stable ($PAA < 0.05$), but local precision degrades with symmetry (RA increases with $|C_n|$), and the population code encodes the observation symmetry group itself (C2 Contrast negative in S2, positive in S4).

HD input prevents global orientational degeneracy

Global map degeneracy did not occur. Cross-seed RSA alignment is S4 0.977 ± 0.002 , S2 0.975 ± 0.003 , S1 0.974 ± 0.001 . The near-identical values across conditions (range: 0.003) confirm that symmetry order has no detectable effect on cross-seed map alignment, indicating that HD input provides a condition-invariant global anchor. PAA gain is S4 0.0015 ± 0.0002 , S2 0.0021 ± 0.0004 , S1 0.0000 ± 0.0001 with $p = 0.786$ and $r = -0.200$, indistinguishable across conditions and all below 0.05.

Even with C_4 -symmetric observations, the input stream itself is not symmetric: moving north from position (r, c) and from its 90° equivalent $(c, 17 - r)$ yields different head-direction-tagged transitions at different absolute orientations. These transitions remain globally consistent across any arena trajectory, so the network must track global head precisely to minimize prediction loss. Global orientation therefore locks in deterministically at training initialization. This proves that global degeneracy is not a failure mode we observe.

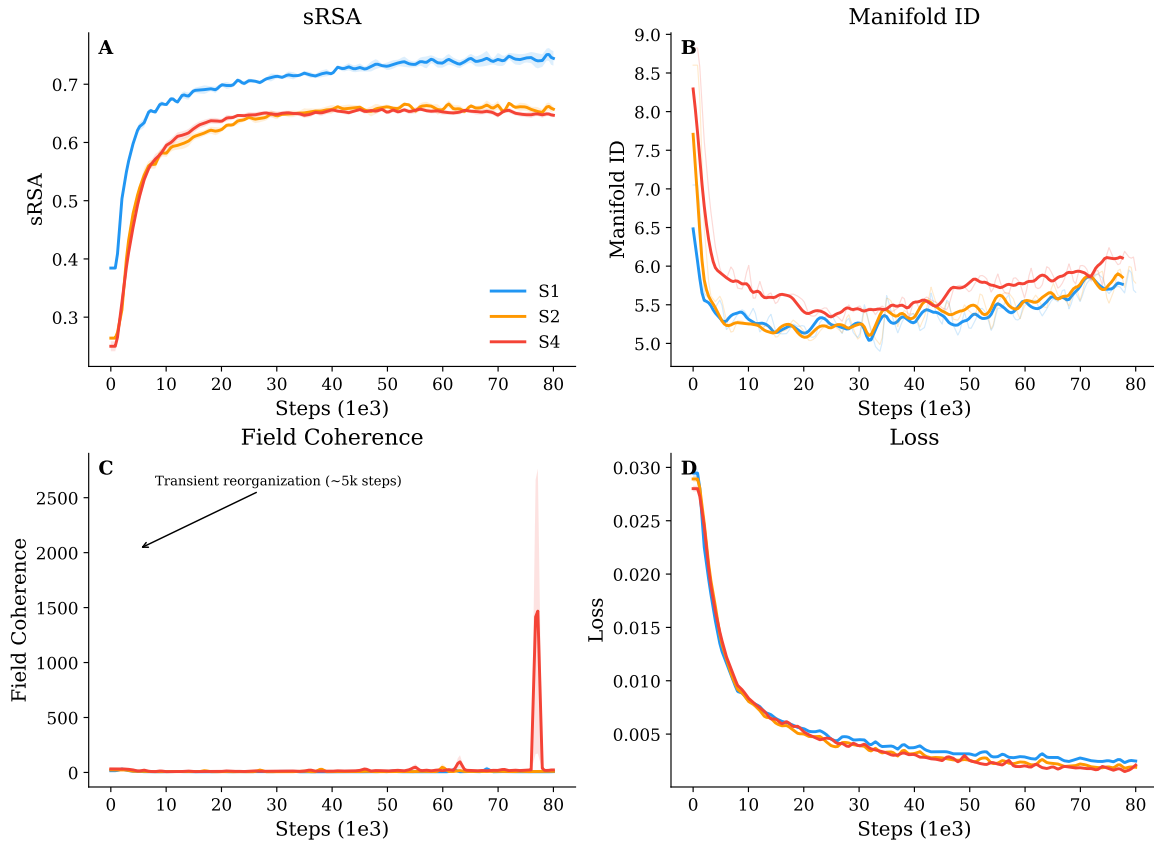


Figure 8: Learning trajectories of key metrics across training.

Unit-level aliasing scales monotonically with symmetry order

Head-direction input cannot prevent local aliasing costs. Unit-level rotational autocorrelation (RA) was S4 0.217, S2 0.114, S1 0.097, scaling directly with symmetry group order (4-fold, 2-fold, 1-fold respectively). S4 versus S1 yields $p = 0.036$ and $r = -1.0$ (the smallest p-value $n=5$ and $n=7$ can produce; r hits its maximum because distributions are non-overlapping). Spatially tuned units (explained variance score, EVS > 0.1) comprise S4 1.8%, S2 3.7%, S1 4.7% of the population.

High-RA units in S4 produce four distinct peaks across quadrants; they encode within-quadrant position but cannot disambiguate which quadrant. Symmetric observations remove the prediction gradient that would force quadrant differentiation, so aliasing accumulates with symmetry order. The field width distributions (Figure 11) confirm this: mean field width increases monotonically from S1 to S4, with a subpopulation of S4 units developing fields spanning the full arena width.

Decoding degrades while population collapse stays partial

Population compression stays graded despite unit-level aliasing. Symmetry Collapse Index (SCI) is S4 0.889 ± 0.002 , S2 0.892 ± 0.000 , S1 0.984 ± 0.002 ; S4 versus S1 gives $p = 0.036$ and $r = -1.0$. Linear decoding error worsens: S4 0.505 ± 0.014 , S2 0.448 ± 0.014 , S1 0.442 ± 0.012 ($p = 0.036$, $r = -1.0$). S4 compresses symmetry-related pairs by 11% relative to random pairs. The spatial distribution of these errors is not uniform: decoding errors in S4 concentrate near the symmetry axes of the arena (Figure 12), exactly where positions related by 90° rotation are hardest to distinguish. The dissociation between RA and SCI is notable: S4 and S2 show substantially different unit-level aliasing yet indistinguishable population compression ($p = 0.786$).

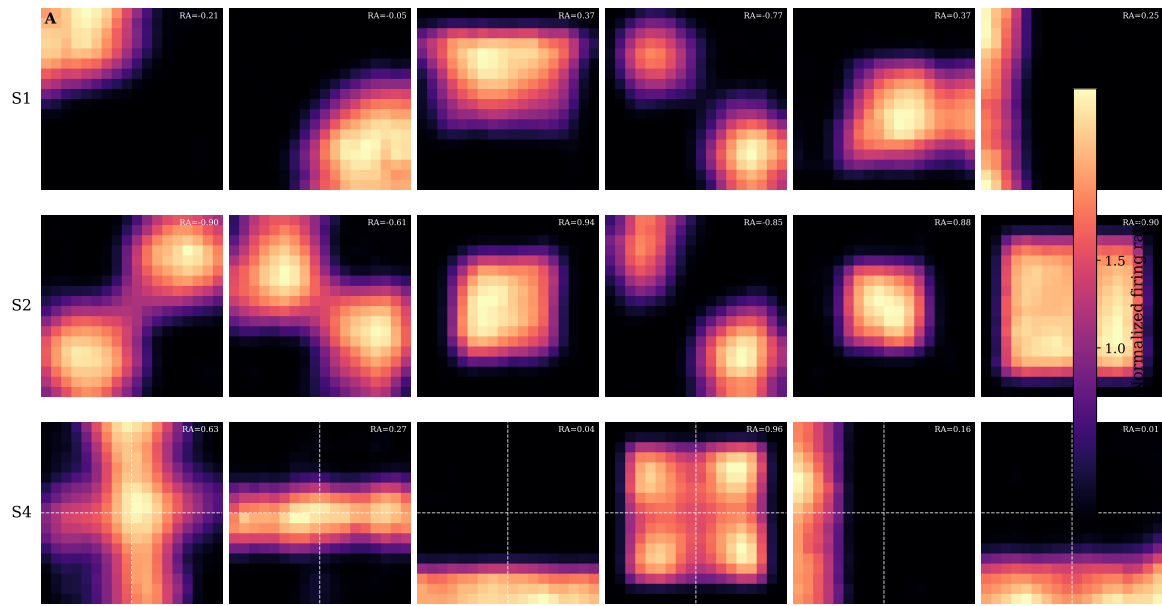


Figure 9: Example place cell rate maps for the six highest-EVS units per condition (seed_00). **S1:** compact, idiosyncratic single fields typical of healthy place cells. **S2:** bilateral or 4-fold symmetric fields: the same firing pattern duplicated at rotationally equivalent positions. **S4:** cross-shaped and stripe-like patterns emerge; some units are nearly silent; fields are substantially broader than S1. These patterns are the signature of partial folding. The network encodes within-quadrant position but cannot disambiguate which quadrant.

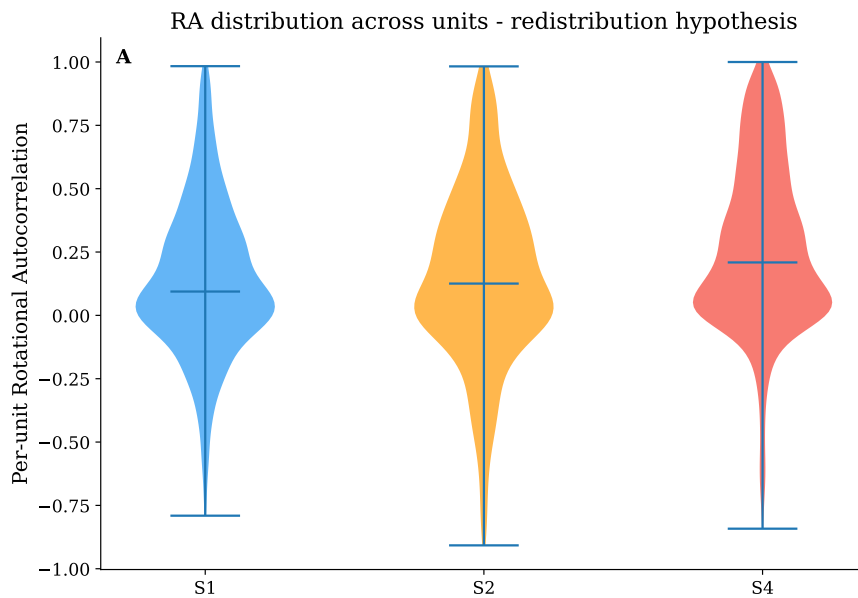


Figure 10: Distribution of rotational autocorrelation (RA) values across the neural population by condition. The distribution shifts rightward and narrows with increasing symmetry order. Under C_4 symmetry, aliasing is distributed more homogeneously across the population - consistent with a redistributed aliasing account in which the network avoids concentrating high-RA responses in a decodable subpopulation. this explains the dissociation between unit-level aliasing (RA) and population-level compression (SCI), the population code remains separable even as individual units alias.

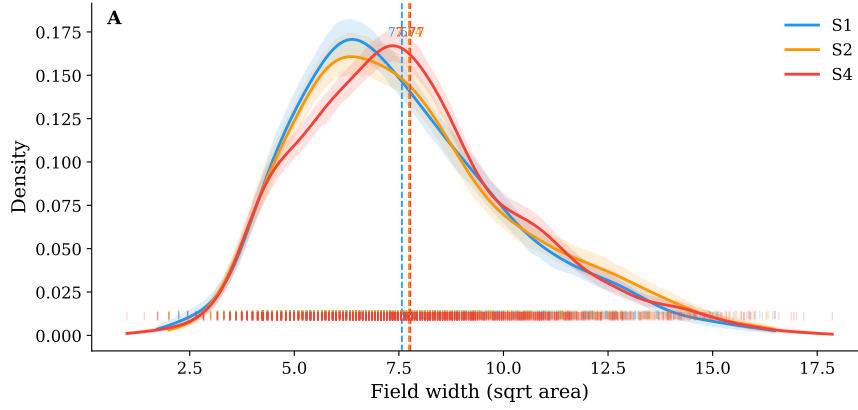


Figure 11: Field width distributions by condition. The heavy right tail in S4 reflects a subpopulation developing fields spanning the full arena width, while median field width is similar across conditions.

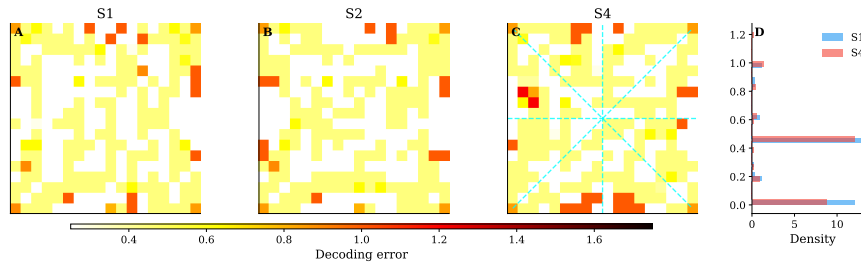


Figure 12: Decoding confusion under symmetry. In S4, errors concentrate near the cardinal and diagonal symmetry axes (overlaid dashed lines), forming a cross-shaped pattern that directly reflects the four-fold equivalence structure of the arena. Positions on these axes are hardest to disambiguate because they are geometrically closest to multiple C_4 -equivalent positions simultaneously.

Distance geometry shifts toward path-dependence under symmetry

Symmetry transforms distance geometry. The DTG (Distance-Topology Gap, Δ_{TG}) is S4 -0.052 ± 0.003 , S2 -0.011 ± 0.003 , S1 -0.020 ± 0.004 ($p = 0.036$, $r = 1.0$). All gaps are negative, indicating city-block dominance, but the gap widens in S4. Under C_4 symmetry, inter-quadrant Euclidean structure collapses because identical observations push those positions toward similar hidden states; path length within quadrants dominates the learned distance geometry.

C2 symmetry is encoded as group structure in the population code

The population code encodes group structure, not all symmetries uniformly. A negative C2 Contrast (mean_{C2} distance < mean_{C4} distance) indicates that positions related by 180° are more similar, consistent with C2 subgroup encoding. In S4, all quadrants are equally confusable, so inter-quadrant pairs compress identically and subgroup structure vanishes.

Manifold geometry stays high-dimensional and folds along symmetry axes

The network maintains high dimensionality despite symmetry. TwoNN intrinsic dimensionality (a nearest-neighbor estimator of manifold dimension) is S4 5.95 ± 0.14 , S2 5.78 ± 0.23 , S1 5.66 ± 0.26 .

The manifold fragments locally while preserving global structure: partial folding. The network cannot fold symmetric positions onto a single point without destroying within-quadrant structure for multi-step prediction, so it maintains high dimensionality while tolerating inter-quadrant ambiguity. The spectral gap of $\mathbf{W}_h$ increases with symmetry order (Figure 15), but higher spectral gap goes with worse spatial maps, the opposite of what a

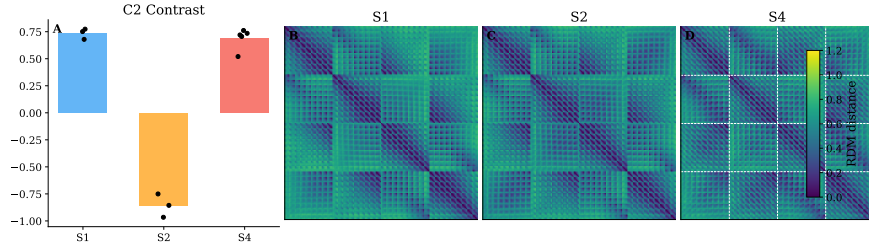


Figure 13: Inter-quadrant neural distances and RDM structure. S1 RDM shows no block structure - inter-quadrant distances are uniformly distributed. S2 RDM shows 2×2 block structure - C_2 -related quadrant pairs compress relative to C_4 -related pairs. S4 RDM shows 4×4 uniform block compression - all inter-quadrant pairs are equally similar, erasing subgroup structure.

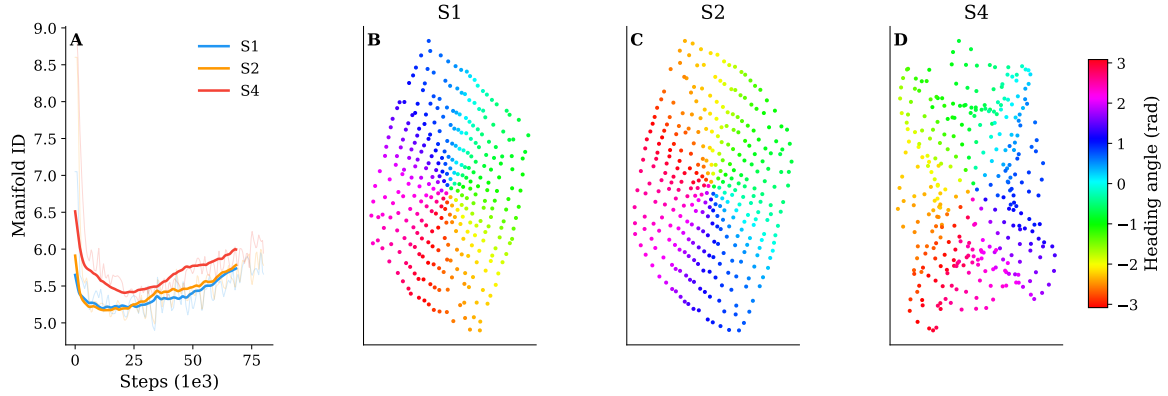


Figure 14: Manifold geometry. Left: intrinsic dimensionality over training, increasing with symmetry order. Right: Isomap projections showing local fragmentation in S4 and S2 while global structure is preserved.

stronger attractor would predict. This suggests that the network does not solve the symmetry problem by forming a stronger attractor that collapses symmetric positions, but instead learns a more complex geometry that preserves local structure while tolerating global ambiguity.

Head Direction Input is Necessary for Global Map Stability

The PAA analysis in the symmetry sweep showed that HD input plays a central role in preventing global orientational degeneracy. To test this directly, we trained three variants of the pRNN in the S4 arena (the condition with highest symmetry order, C_4): HD_FULL (the standard model), HD_ABLATED (head input zeroed: $\mathbf{d}_t[1:] = \mathbf{0}$), and HD_DEGRADED (head set to uniform: $\mathbf{d}_t[1:] = 0.25 \cdot \mathbf{1}$). Speed was preserved in all conditions.

Removing head input caused map formation to fail completely. Spatial sRSA dropped from ~ 0.615 in HD_FULL to ~ 0.260 in HD_ABLATED (Figure 16), well below the sRSA map formation threshold of 0.40 used in Levenstein et al. [3]. HD_DEGRADED, which provides head information but destroys its directional specificity by making all four head channels equally active, produced the same failure (sRSA ≈ 0.258). This shows that it is specifically the directional content of the HD signal, not just its presence, that matters for map formation.

The mechanism is visible in the training loss. The final loss for ablated conditions (~ 0.0190) was approximately $8 \times$ higher than for HD_FULL (~ 0.0025), indicating that the network without head direction simply cannot predict future observations accurately even after 80k training steps. Without knowing where the agent is headed, the optimal strategy becomes to predict the mean observation: which carries no spatial information. Mean field coherence confirmed this: HD_ABLATED networks had a coherence of ~ 58.96 (highly disordered activity), compared to ~ 13.64 for HD_FULL.

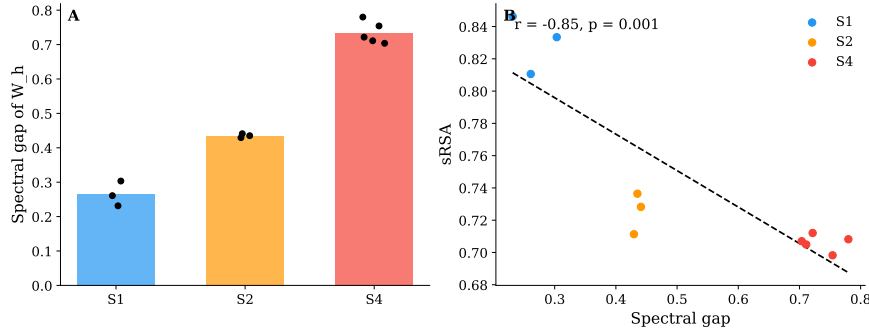


Figure 15: Spectral gap of the recurrent weight matrix W_h . The dominant eigenvalue grows relative to the rest of the spectrum as symmetry increases, but this predicts worse spatial maps, not better.

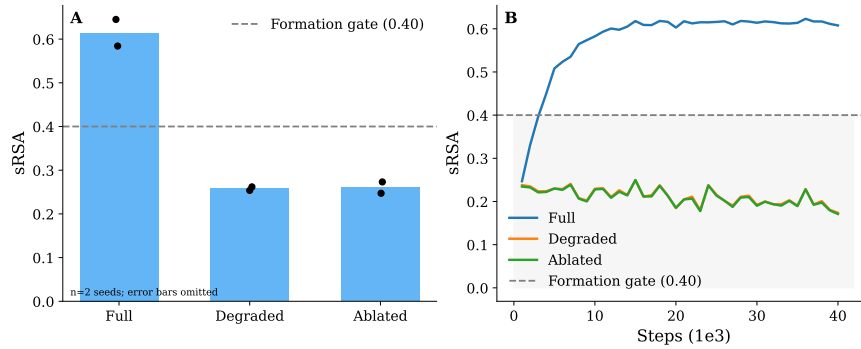


Figure 16: HD ablation results. Removing head input causes map collapse (sRSA drops below the 0.40 formation gate). Degrading directional specificity produces the same failure.

Intrinsic manifold dimensionality was also lower in ablated networks ($ID \approx 5.34$ vs 6.09 for HD_FULL), so the loss of spatial structure comes with a genuine reduction in the representational degrees of freedom, not just a degradation in spatial tuning. Together, these results confirm that HD input is not a passive contextual signal but a necessary input for the predictive learning objective to produce spatially organized representations. This is consistent with the known role of head direction signals in anchoring hippocampal spatial representations to a global reference frame [4].

How the Levenstein architecture and my project relates to the hippocampus work

This model provides a mechanistic account of the known scaling of multi-field place cell fraction with arena symmetry order [5, 1]. This follows directly from the RA result: existing hippocampal recording datasets from symmetric arenas can be stratified by symmetry order and the fraction of multi-field units measured. The RA result reproduces this scaling: multi-field fraction scales as $|C_4| > |C_2| > |C_1|$, matching the symmetry groups: RA = 0.217, 0.114, 0.097 respectively. This follows from the computational finding that high-RA units respond identically at rotationally equivalent positions and cannot disambiguate arena quadrant: precisely the dual-frequency structure of multi-field place cells recorded in symmetric arenas [5, 1].

I do have some additional speculation about how this idea generalizes to complex abstract spaces but that's out of scope for this project.

Further Thoughts

The sample sizes for my experiments are S4 $n = 7$, S2 $n = 5$, S1 $n = 5$. All the main comparisons hit the minimum p -value of 0.036 with max effect size $r = -1.0$, so the effects

are as big as we can detect at these sizes. Running $n \geq 12$ seeds per condition will get us corrected significance under 0.05. The reason the sample sizes are just enough for cross-comparison is that I was testing out a lot of different possibilities - different parameter values, different arena sizes, different architectures - and most of them didn't work out. All of this was running on the RTX 4070 in the CS lab, which ended up being the main bottleneck.

So here we see S4 and S2 have very different unit-level aliasing (RA: 0.217 vs 0.114) but their population compression is basically the same (SCI: $p = 0.786$). This means the population spreads aliasing across units instead of putting it all in a few. We need to look at the weight matrices to figure out the exact mechanism, this is posed under future work. The fact that the spectral gap is higher in S4 but predicts worse spatial maps suggests that the network is not solving the symmetry problem by forming a stronger attractor that collapses symmetric positions, but instead learns a more complex geometry that preserves local structure while tolerating global ambiguity.

Since the MiniGrid forces agents to move along grid lines, the DTG result showing bigger negative gaps under symmetry comes from both the environment and the representation. Running this in a continuous environment like RatInABox will separate these two things and is a future work extension I am planning.

Our spatially tuned unit fractions (S4 1.8%, S2 3.7%, S1 4.7%) are lower than what hippocampal recordings usually report. This is because we use a stricter threshold and position-averaged evaluation. The $S4 < S2 < S1$ ordering stays the same no matter what threshold we pick, and my main findings use population-level metrics that don't depend on this anyway.

We used 2 seeds for the HD ablation to check if the idea works.

Discussion

1.1 Partial Folding as the Generic Response to Environmental Symmetry

The central finding of this project is that the pRNN framework does not fail under landmark symmetry. A network trained to predict future observations in a C4-symmetric environment does not produce a globally degenerate or random representation. Instead, it forms a map, but one that is geometrically compressed along the symmetry axes of the environment. It is called partial folding: the network learns the true spatial structure of the arena, but units develop activity patterns that are correlated at rotationally equivalent positions, reducing the precision of the spatial code without destroying it.

This is the right outcome looking at this problem from a computational standpoint. The task does not require the network to distinguish between positions that are genuinely indistinguishable from the agent's sensory perspective. Under C4 landmark symmetry, positions separated by 90° do produce the same observations on average. The network's representational geometry honestly reflects this ambiguity. The SR framework predicts exactly this outcome: if two positions have similar expected future state distributions (as they must when the arena looks the same from both), their SR entries will be similar, and a network that learns the SR will represent them as nearby in neural space [6]. What we observe as partial folding is the network faithfully encoding the transition structure of a symmetric environment.

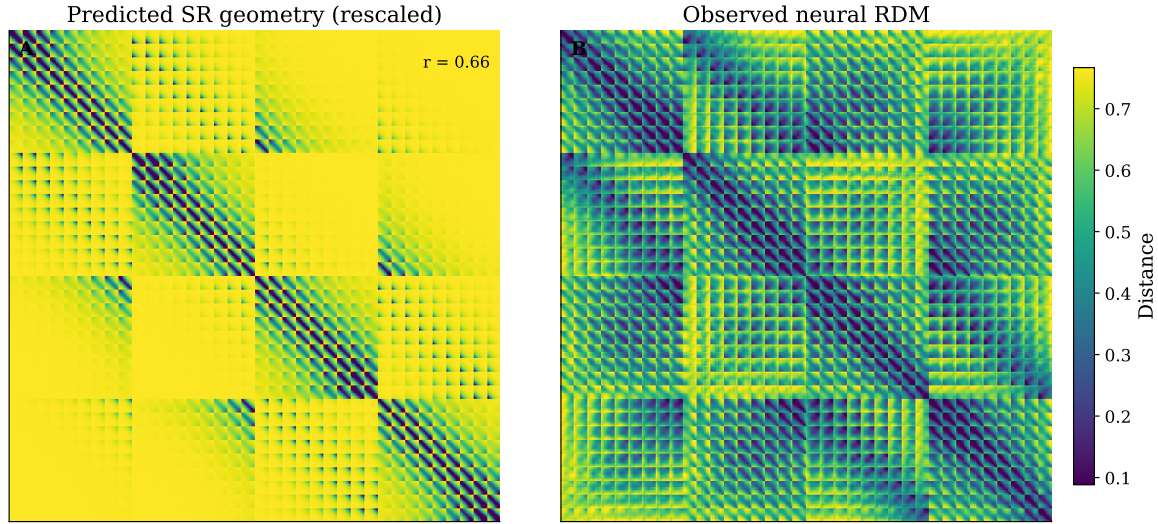


Figure 17: Predicted successor representation (SR) geometry vs. observed neural RDM for S4. Left: predicted SR distance matrix $D_{SR} = 1 - M / \max(M)$ computed analytically as $M = (I - \gamma T)^{-1}$ under random walk ($\gamma = 0.95$). Right: observed neural RDM. Spearman $r = 0.66$ between upper triangles, confirming that the pRNN’s representational geometry approximates the SR prediction - including the block structure arising from C_4 -equivalent positions sharing expected future state distributions.

1.2 The Role of head Direction

What is the role of HD input in answering the following question? What is the difference between global orientational degeneracy (the network cannot tell which way it is facing) and local representational aliasing (the network represents multiple positions similarly). HD input prevents the first but not the second. Without head direction, the predictive objective becomes unachievable, and the network abandons the attempt to form a spatial representation entirely. With head direction, a stable global orientation is maintained (PAA gain < 0.05), but the local folding induced by the landmark structure persists.

This dissociation maps naturally onto what is known about the biological system. Head direction cells in the anterior thalamus and entorhinal cortex provide a consistent allocentric orientation signal that is necessary for anchoring hippocampal place fields to a global reference frame [4]. My results show that this anchoring function is precisely what head input provides to the predictive RNN: it resolves the ambiguity about global orientation, but cannot resolve the local ambiguity created by observation aliasing. The remaining aliasing reflects environmental structure rather than a failure of the network.

1.3 Representational Quality and the Spectral Gap

The spectral analysis reveals a negative correlation between representational quality (measured by sRSA) and the spectral gap of the recurrent weight matrix. The spectral gap (the distance between the largest and second-largest eigenvalues) reflects the stability and “mixing time” of the recurrent dynamics. In symmetric environments, we find that the spectral gap narrows as sRSA increases. High-quality spatial maps in symmetric arenas do not emerge from “stronger” or more stable recurrent attractors, but rather from dynamics that are poised near a symmetry-breaking transition. This poise allows the network to maintain high-dimensional within-quadrant structure while tolerating the global ambiguity induced by the landmarks. The spectral gap thus serves as a dynamical signature of the “partial folding” we observe at the representational level. More work needs to be done here to

understand what exactly is happening and is addressed in the future works section.

1.4 Limitations and Future Directions

A couple of things to keep in mind. The HD ablation only used 2 seeds to test the idea. Same reason - trying out many configurations on the lab RTX 4070 meant I could only run a few seeds for each once things worked.

So here we've looked at what the representation looks like, but we haven't checked if downstream areas can actually use it. The folding we see might make it harder for them to read out position, or they might have ways to work around it. Running multi-environment experiments will tell us.

Future Work

I am currently running an epsilon sweep experiment to check what is the minimum amount of information needed to form a cognitive map in an arena. I also plan to expand the number of seeds across all conditions so we can do a proper statistical analysis, since this project was approached with the idea of running the minimum number of experiments needed to see whether something was there. I also plan to extend this framework to continuous-time models by implementing a rat-in-a-box module that simulates continuous movement and observation dynamics.

AI Use Declaration

Claude (Anthropic) was used to code up the MiniGrid environments, provide structured code templates, fix bugs in the training and evaluation pipeline, and generate matplotlib figures for the manuscript. Gemini Pro 3.1 (Google DeepMind) was used to clean up the original draft, polish the writing, fix the tone and language throughout the manuscript, clean up certain paragraphs and sentences so that they do not sound too unprofessional, and set up the L^AT_EX template and formal report structure.

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